

Greedy algorithms

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Greedy Algorithms

- Optimization problems are problems that have several solutions and we are to choose the solution that is optimal (maximum/minimum)
- A greedy algorithm has the greedy choice property in all of its decisions
 - **Greedy choice:** always makes the choice that looks best at the moment.
- Requirements:
 - Optimal sub-structure that can be exploited using a recursive algorithm
 - optimal solution to the problem contains within it optimal solutions to subproblems
 - Greedy choice property leads to optimal solution
- Examples:
 - Prim's and Kruskal's algorithm for finding MST
 - Dijkstra's algorithm for finding single source shortest path algorithm
 - Huffman codes for data compression

Huffman Codes

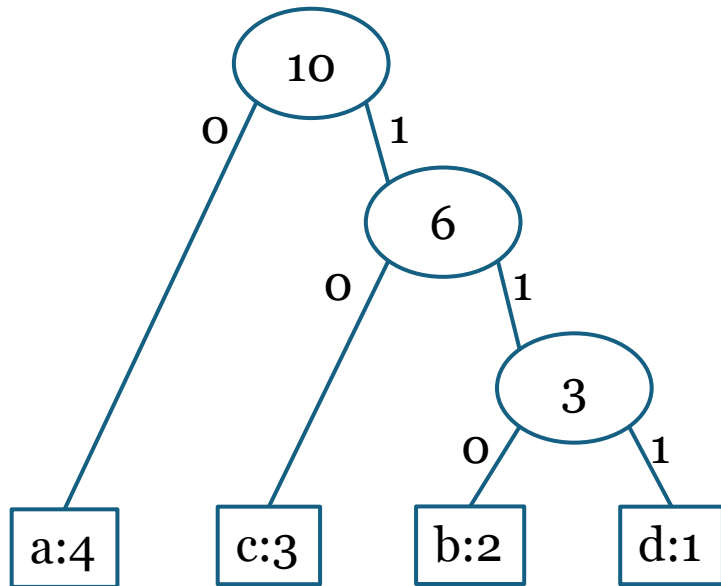
- Huffman codes compress data where instead of using a fixed length binary code like (ASCII), a variable length code is used
 - Length of the code depends on the frequency of the characters
 - Higher frequency characters have shorter codes and lower frequency characters have longer codes
- For example: *aaaabbcccd*

In ASCII: 01100001 01100001 01100001 01100001 01100010
01100010 01100011 01100011 01100011 01100100

Using Huffman code: 0 0 0 0 110 110 10 10 10 111

Huffman code generation

- Generating the optimal Huffman codes is done using a greedy algorithm
- Generates a table of how frequently the characters occur
- Builds a tree of prefix codes as follows:



Character	Frequency	Code
a	4	0
b	2	110
c	3	10
d	1	111

Encryption and Decryption

- After generating the codes, the characters are encoded using the variable length codes
- However, since the codes are dependent on the frequency of the characters, information about the character codes is needed for decoding such a file
 - i.e. which code corresponds to which character – a part of a file header
 - For example, we can explicitly write the preorder traversal of the Huffman tree : acbd###NUL
 - Here NUL is used to indicate end of header
- While reading the file, the tree is reconstructed from the header to obtain the codes for each character.
- The file is read using the codes